Directional transitions in an accelerated gradient restart system

Abstract

We study the accepted restart states of a specified accelerated gradient method on a two-parameter family of strongly convex quadratic functions in $\mathbb{R}^2$. The method uses the strict gradient-angle test and the reject-and-recompute rule of Bao, Chen, Li and Shen. Although every accepted point sequence converges geometrically to the minimizer, its normalized restart states have three different regimes. They converge to an eigendirection, form a two-cycle, or enter a regime with periodic directions or two closed arcs as their limit set. We derive the exact return map from the original iteration, including its switching equalities. In the third regime the restart intervals are 3 and 4, and their frequencies are given by a logarithmic rotation. For each fixed second eigenvalue, every frequency in $(0, 1)$ is realized by exactly one parameter. An explicit rational quadratic gives an irrational frequency and a non-eventually-periodic restart word. The rotation arguments are classical; the result is their exact realization and parameter classification for the stated restart rule.

Keywords: Discrete dynamical system; gradient restart; normalized orbit; rotation coding

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1 Introduction

Restarting changes both the trajectory of an accelerated method and the times at which its momentum is removed. The accepted points may approach a minimizer while their directions and restart intervals continue to vary. To describe this behavior one must keep the event rule, the accepted replacement point and the reset clock in the dynamical state.

O'Donoghue and Candès [1] introduced the gradient-angle restart heuristic and related it to modal oscillation. Bao, Chen, Li and Shen [2] give a precise reject-and-recompute implementation and prove convergence results for the original points. Their stronger bounds under uniformly bounded restart intervals do not determine the resulting direction sequence or the exact restart word. Implementation conventions matter here. For example, Moursi, Pavlovic and Vavasis [3, Remark 2 and Algorithm 1] retain the triggering trial in their analyzed scheme. That assignment gives a different return map from the replacement step considered below.

Normalized dynamics of convergent quadratic solvers has a substantial history. Pronzato, Wynn and Zhigljavsky [4] study two-point limits for a family of weighted gradient methods; their Remark 3 also discusses more complicated normalized behavior under relaxation. Van den Doel and Ascher [5] examine parameter-dependent normalized dynamics, including planar ratio maps, while the original errors decrease. Thus the coexistence

of point convergence and nonconvergent directions is established background. Our purpose is to classify the directions and event clocks of one particular restarted accelerated system on an explicit spectral region. We use classical rotation coding after deriving its exact occurrence in the algorithm.

We first specify the recurrence. Fix

$$\frac{1}{7} < v < \frac{1}{5} < u < \frac{3}{7}, \qquad H = \operatorname{diag}(1-u, 1-v), \qquad f(x) = \frac{1}{2} x^{\mathsf{T}} H x. \tag{1}$$

Use step size 1, initial values $x_0 = y_0 = (\xi, \zeta)$ with $\xi > 0$ and $\zeta \neq 0$, and local clock $j = 1$. The rational schedule $t_j = (j+1)/2$ gives the momentum coefficient

$$\beta_j = \frac{t_j - 1}{t_{j+1}} = \frac{j-1}{j+2}.$$

At global iteration $k \geq 1$, form the trial, provisional extrapolate and test

$$\begin{aligned} z_k &= (I - H) y_{k-1}, \\ \widehat{y}_k &= z_k + \beta_j (z_k - x_{k-1}), \\ c_k &= \langle z_k - x_{k-1},\, y_{k-1} - z_k \rangle. \end{aligned} \tag{2}$$

If $c_k > 0$, discard z_k and set

$$x_k = (I - H) x_{k-1}, \qquad y_k = x_k, \qquad j = 1. \tag{3}$$

If $c_k \leq 0$, set $x_k = z_k$, $y_k = \widehat{y}_k$ and increment j by one. In particular, equality does not trigger a restart, and the replacement step in (3) leaves the clock at 1. These are the assignments in Algorithm 2 of the versioned preprint [2], with its permitted rational schedule parameter $r = 2$. The step is strictly admissible: the exact curvature constants are $\mu = 1 - u$ and $L = 1 - v < 1$.

Let $\tau_0 = 0$, and let τ_n, $n \geq 1$, denote the global index of the nth execution of (3). We write

$$E_n = x_{\tau_n}, \qquad d_n = \tau_{n+1} - \tau_n, \qquad r_n = \left| \frac{E_{n,2}}{E_{n,1}} \right|, \qquad S_n = \frac{E_n}{\|E_n\|}.$$

Existence of all these quantities is part of the following theorem. The state E_0 is an initial block boundary, not an actual restart event. Put

$$A = \left(\frac{v}{u}\right)^3, \qquad B = A \frac{1 - 5v}{5u - 1}, \qquad R^2 = \frac{u^3 (1-u)^2 (5u-1)}{v^3 (1-v)^2 (1-5v)}, \quad R > 0. \tag{4}$$

The ω-limit of a sequence means its set of subsequential limits.

Theorem 1. *For every parameter pair and initialization specified above, the recurrence is defined for all k. Every accepted iterate has a positive first coordinate and a nonzero second coordinate, every d_n is 3 or 4, and*

$$\|x_k\| \leq u^{3\lfloor k/4 \rfloor} \|x_0\| \qquad (k \geq 0). \tag{5}$$

The normalized restart states satisfy the following classification.

1. *If $B < 1$, the word (d_n) is eventually constantly 4, and $S_n \to (1, 0)$.*

2. *If $B = 1$, the word is eventually constantly 4. For some $r_* > 0$ depending on the initialization, the normalized restart states eventually alternate between*

$$\frac{(1, r_*)}{\sqrt{1 + r_*^2}} \quad \text{and} \quad \frac{(1, -r_*)}{\sqrt{1 + r_*^2}}.$$

Their signed and projective least periods are both 2.

3. *If $B > 1$, every ratio sequence enters $(AR, BR]$ after finitely many blocks. Define*

$$a = -\log A, \qquad b = \log B, \qquad \ell = a + b, \qquad \alpha = \frac{a}{\ell} \in (0, 1). \tag{6}$$

If $N_4(N)$ counts the occurrences of 4 in $d_0, \ldots, d_{N-1}$, then

$$\frac{N_4(N)}{N} \longrightarrow \alpha, \qquad \frac{N - N_4(N)}{N} \longrightarrow 1 - \alpha.$$

If α is irrational, the word is not eventually periodic, and the signed ω-limit is exactly $\Gamma_+ \cup \Gamma_-$, where

$$\Gamma_\pm = \left\{ \frac{(1, \pm r)}{\sqrt{1 + r^2}} : AR \le r \le BR \right\}. \tag{7}$$

The projective ω-limit is the image of these arcs in $\mathbb{RP}^1$ and is infinite.

If $\alpha = p/q$ in lowest terms, $1 \le p < q$, then after a finite prefix both the absolute ratio and the restart word have least period q. Each word period contains p entries equal to 4. The signed and projective normalized states have least period q when p is even and $2q$ when p is odd. Their ω-limit is the resulting finite orbit.

All assertions include initial states and later states at which $r_n = R$, AR or BR occurs.

The theorem concerns restart directions. It does not identify the ω-limit of all normalized accepted points between restarts. A projective direction identifies a vector with its negative; it does not identify $(1, r)$ with $(1, -r)$ for $r > 0$. The arcs in (7) are physical limit sets. The circle used in the proof is a logarithmic quotient, not a geometric invariant circle in the original state space.

The parameter separating the regimes is also explicit up to one strictly monotone scalar equation.

Theorem 2. *For each $v \in (1/7, 1/5)$ there is a unique $u_*(v) \in (1/5, 3/7)$ satisfying*

$$u_*^3(5u_* - 1) = v^3(1 - 5v). \tag{8}$$

The regimes $B > 1$, $B = 1$ and $B < 1$ correspond respectively to $u < u_*$, $u = u_*$ and $u > u_*$. On $(1/5, u_*)$, the function $u \mapsto \alpha(u, v)$ is a continuously differentiable, strictly increasing bijection onto $(0, 1)$. Rational-frequency parameters are countable and dense, while irrational-frequency parameters are dense and have full one-dimensional Lebesgue measure.

The assertions are restricted to (1), the stated schedule and non-axis initializations. Rational frequencies are dense in the rotation regime, so the irrational conclusion is not an open-parameter robustness statement. No claim about excluded spectral endpoints or other restart assignments is made. Our source comparison concerns the specified versions in the references. In particular, [5] is the inspected July 2011 author manuscript; equivalence to its final publisher version has not been checked. The contribution claimed here is the exact classification of this recurrence, without a general priority assertion about normalized gradient dynamics.

We prove Theorem 1 in Section 2 and Theorem 2 in Section 3. Section 4 gives a rational quadratic for which the restart frequency is irrational.

2 The return map and its normalized dynamics

We first derive the event map from the accepted and rejected trials. This step is needed before passing to any projective coordinate.

Lemma 1. *At a block boundary $E = (E_1, E_2)$ with $E_1 > 0$, $E_2 \neq 0$, the first trigger occurs at local step 3 if $|E_2/E_1| > R$ and at local step 4 otherwise. The accepted return is*

$$E' = \begin{cases} \operatorname{diag}(u^3, v^3)E, & |E_2/E_1| > R, \\ \operatorname{diag}\big(u^3(5u-1)/4,\, v^3(5v-1)/4\big)E, & |E_2/E_1| \le R. \end{cases} \tag{9}$$

The first coordinate remains positive, both coordinates remain nonzero, and the second-coordinate sign changes between returns exactly on the length-4 branch.

Proof. Write h for either u or v. At a reset, the accepted point and the extrapolate both equal E and the clock equals 1. Let $P_j(h)$ be the coefficient of the jth trial, assuming that earlier trials in the block have not triggered. Then

$$P_0(h) = 1, \quad P_1(h) = h, \quad P_{j+1}(h) = h\big((1+\beta_j)P_j(h) - \beta_j P_{j-1}(h)\big).$$

The first four trial polynomials are

$$P_1 = h, \qquad P_2 = h^2, \qquad P_3 = \frac{h^2(5h-1)}{4}, \qquad P_4 = \frac{h^3(7h-3)}{4}. \tag{10}$$

Before the jth trial, the corresponding extrapolate coefficient is $P_j(h)/h$. Hence the coefficient multiplying the square of the initial coordinate in the test is

$$C_j(h) = (P_j(h) - P_{j-1}(h))\frac{(1-h)P_j(h)}{h}.$$

Substitution gives

$$\begin{aligned} C_1(h) &= -(1-h)^2, \qquad C_2(h) = -h^2(1-h)^2, \\ C_3(h) &= -\frac{5}{16}h^3(1-h)^2(5h-1), \\ C_4(h) &= -\frac{1}{16}h^4(1-h)^2(7h-3)(7h-1). \end{aligned} \tag{11}$$

The actual test is $C_j(u)E_1^2 + C_j(v)E_2^2$. Its first two values are negative. At the third trial, $C_3(u) < 0 < C_3(v)$, so positivity is equivalent to $|E_2/E_1| > R$, with R as in (4). On this branch the trial P_3 is discarded. The replacement is $hP_2 = h^3$, which proves the first line of (9).

If the third test is nonpositive, its trial is accepted. This includes equality. Since $1/7 < h < 3/7$, the fourth coefficient in (11) is strictly positive for both coordinates. Thus the fourth trial triggers, and its replacement is $hP_3 = h^3(5h-1)/4$. This proves the second line.

All accepted coefficients are nonzero. Those in the first coordinate are positive because $u > 1/5$. At a length-3 return both factors are positive, while at a length-4 return the second is negative because $v < 1/5$. The same conditions therefore hold at the next block boundary. They also show positivity and nonvanishing at each accepted point inside the block. Notice that in a length-4 block the second-coordinate sign can already change at the accepted third trial; the assertion here concerns the signs at successive restart states. ∎

Proposition 2.1. All restart events exist, the estimate (5) holds, and a literal block of length $d \in \{3, 4\}$ uses $d + 1$ original-gradient evaluations.

Proof. The initial point satisfies the hypotheses of Lemma 1. Induction gives a next event in at most four steps, with the same nonzero-coordinate conditions. This proves indefinite well-definedness and existence of all events.

For $h \in (1/7, 3/7)$,

$$\frac{|5h-1|}{4} < \frac{2}{7} < 1.$$

Both return matrices in (9) have norm at most u^3, because $0 < v < u < 1$. Thus $\|E_n\| \le u^{3n}\|x_0\|$. The accepted intermediate points use P_1, P_2, and, if reached, P_3; each of these diagonal maps has norm at most 1. If $\tau_n \le k < \tau_{n+1}$, then

$$k < \tau_{n+1} \le 4(n+1), \qquad \lfloor k/4 \rfloor \le n.$$

It follows that $\|x_k\| \le \|E_n\| \le u^{3\lfloor k/4 \rfloor}\|x_0\|$, including $k = 0$ and event indices.

Every global step forms one trial gradient. A triggering step also forms the replacement gradient at the preceding accepted point. A block contains d trials and one replacement, hence costs $d + 1$ calls. Initialization uses no gradient. This counts the literal recurrence without exploiting a special quadratic shortcut; the discarded trial is not an accepted iterate. ∎

We now study the ratio process. Lemma 1 gives the exact rule

$$r_{n+1} = \begin{cases} Ar_n, & r_n > R, \\ Br_n, & 0 < r_n \le R, \end{cases} \qquad d_n = \begin{cases} 3, & r_n > R, \\ 4, & r_n \le R. \end{cases} \tag{12}$$

Here $0 < A < 1$ and $B > 0$.

Proof of Theorem 1. The preliminary assertions and radial bound follow from Lemma 1 and Proposition 2.1. We divide the direction argument according to B.

Case 1: $B < 1$**.** If $r_0 > R$, consecutive short blocks multiply the ratio by A until it first enters $(0, R]$. This takes finitely many steps. Thereafter the ratio remains in that interval, every block has length 4, and it is multiplied by B each time. Hence $r_n \to 0$. Although the second-coordinate sign alternates between returns, its normalized magnitude tends to zero, while the first coordinate is positive. Therefore $S_n \to (1, 0)$.

Case 2: $B = 1$**.** The same finite-entry argument applies. Once $r_n \le R$, the ratio is a fixed positive number r_* and every block has length 4. The sign rule gives the asserted two-cycle. Its points are distinct and are not antipodal, since their first coordinates are positive. They therefore have distinct projective images, so both least periods are 2. Equality $r_* = R$ causes no exception.

Case 3: $B > 1$**.** Put $I = (AR, BR]$. If $r_0 > BR$, its consecutive short steps multiply by A until the first value at most BR. This value exceeds $ABR > AR$, so it lies in I. If $r_0 \le AR$, consecutive long steps multiply by B until the first value greater than AR. That value is at most $ABR < BR$, so it also lies in I. Both entry times are finite. Every other initial ratio already lies in I.

The interval is invariant: its part $(AR, R]$ maps to $(ABR, BR]$, and its part $(R, BR]$ maps to $(AR, ABR]$. In particular,

$$R \mapsto BR \mapsto ABR, \qquad AR \mapsto ABR.$$

The first chain lies in I; the second accounts for the excluded lower endpoint if it occurs during the initial prefix.

After this finite entry, relabel the block index from zero and define

$$\theta_n = \frac{\log(r_n/R) + a}{\ell} \in (0, 1], \qquad \iota_n = \begin{cases} 1, & d_n = 4, \\ 0, & d_n = 3. \end{cases}$$

Equation (12), with its strict and nonstrict branches, becomes

$$\iota_n = \mathbf{1}_{\{\theta_n \le \alpha\}}, \qquad \theta_{n+1} = \theta_n - \alpha + \iota_n. \tag{13}$$

This is rotation by $-\alpha$ modulo 1, represented on $(0, 1]$. It includes $\theta_n = \alpha \mapsto 1$ and $\theta_n = 1 \mapsto 1 - \alpha$. Summing gives

$$\sum_{n=0}^{N-1} \iota_n - N\alpha = \theta_N - \theta_0. \tag{14}$$

The right-hand side has absolute value less than 1. The finite pre-entry prefix only adds a fixed constant to this discrepancy, so the two frequency limits hold from the original initialization.

For the sign, let $\sigma_n = 0$ when $E_{n,2} > 0$ and $\sigma_n = 1$ when $E_{n,2} < 0$. Then $\sigma_{n+1} = \sigma_n + \iota_n$ modulo 2. Combining this with (13), the coordinate

$$w_n = \theta_n + \sigma_n \pmod 2 \quad \text{satisfies} \quad w_{n+1} = w_n - \alpha \pmod 2. \tag{15}$$

This is an exact lift, not merely a relation between absolute ratios. Each circle point has a unique representative in $(0, 2]$. Its part $(0, 1]$ has positive second-coordinate sign; its part $(1, 2]$ has negative sign. Recovering θ on the corresponding part gives

$$r = R\exp(\ell\theta - a), \qquad S = \frac{(1, (-1)^\sigma r)}{\sqrt{1 + r^2}}. \tag{16}$$

At a threshold, the half-open convention still gives this same rule. For example, $\theta = \alpha$, $\sigma = 0$ maps to $\theta' = 1$, $\sigma' = 1$, whose representative $w' = 2$ is the same point as 0 modulo 2.

Suppose first that α is irrational. Every forward tail of the circle rotation (15) is dense. We recall an elementary justification. For an irrational step γ on a circle of circumference 1, the pigeonhole principle applied to $0, \gamma, \ldots, M\gamma$ gives $k \geq 1$ for which $0 < \delta = \operatorname{dist}(k\gamma, \mathbb{Z}) \leq 1/M$. The points $0, k\gamma, \ldots, \lfloor 1/\delta \rfloor k\gamma$ form a mesh of gap at most δ, advancing by δ in one of the two orientations. They are forward orbit points. Letting M increase proves density, and translating the starting point proves density of every tail. Apply this fact after rescaling (15) by 2.

On the interior of either sign part, (16) is continuous. Density therefore gives every point of (7) with $AR < r < BR$ as a subsequential limit. To obtain the endpoints, approach $w = 0$ from above, $w = 1$ from below, $w = 1$ from above, and $w = 2$ from below, respectively. These four approaches give

$$\frac{(1, AR)}{\sqrt{1 + (AR)^2}}, \quad \frac{(1, BR)}{\sqrt{1 + (BR)^2}}, \quad \frac{(1, -AR)}{\sqrt{1 + (AR)^2}}, \quad \frac{(1, -BR)}{\sqrt{1 + (BR)^2}}.$$

Every tail meets the required one-sided neighborhoods, so the subsequence indices can be chosen strictly increasing. Conversely, all post-entry states lie on the corresponding half-open arcs, whose compact closure is precisely $\Gamma_+ \cup \Gamma_-$. This proves the exact signed ω-limit, even when the orbit itself hits a threshold or cut point. An eventually periodic word has rational limiting symbol frequencies, so (14) rules it out here.

The map from these compact arcs to $\mathbb{RP}^1$ is continuous and injective, because the first coordinate is strictly positive. Compactness and subsequence extraction show that the projective ω-limit is exactly their image, which is infinite.

Finally let $\alpha = p/q$ in lowest terms. Rotation (13) has least period q, and the representative $\theta \in (0, 1]$ determines r uniquely. Thus the absolute ratio has least period q. The word is q-periodic, and (14) over one period gives exactly p long blocks. If its least period were $m < q$, its frequency would equal an integer divided by m, contradicting that p/q is reduced. Hence its least period is q.

The signed lift returns after m steps exactly when $mp/q \in 2\mathbb{Z}$. This first occurs at $m = q$ if p is even and at $m = 2q$ if p is odd. The half-open representative and (16) give an injective correspondence with signed states, so these are their actual least periods. Positive first coordinates give the same projective periods. The finite periodic orbit is its own ω-limit. All formulas retain the equality branch in (13), completing the proof. ∎

3 The transition parameter

Proof of Theorem 2. Fix $v \in (1/7, 1/5)$, and write

$$K = v^3(1 - 5v) > 0, \qquad g(u) = u^3(5u - 1), \qquad B = \frac{K}{g(u)}.$$

For $u > 1/5$, $g'(u) = u^2(20u - 3) > 0$. Also $g(1/5) = 0$ and

$$K < v^3 < \frac{1}{125} < \frac{216}{2401} = g(3/7).$$

The intermediate value theorem and strict monotonicity give exactly one solution of (8). They also give the stated ordering of the three regimes.

In the interval $1/5 < u < u_*$,

$$a = 3\log(u/v) > 0, \qquad \ell = \log\frac{1 - 5v}{5u - 1} = a + \log B > a,$$

and

$$\frac{\partial\alpha}{\partial u} = \frac{(3/u)\ell + 5a/(5u - 1)}{\ell^2} > 0. \tag{17}$$

Thus α is continuously differentiable and strictly increasing. As $u \downarrow 1/5$, its numerator a tends to the finite positive number $3\log(1/(5v))$, while $\ell \to \infty$. Hence $\alpha \to 0$. As $u \uparrow u_*$, $\log B \to 0$ and $\ell \to a > 0$, so $\alpha \to 1$. This proves the bijection.

The inverse images of rational and irrational frequencies are both dense by continuity and strict monotonicity. The former set is countable by injectivity, hence has Lebesgue measure zero. Its complement has full measure. No assertion that an arbitrary homeomorphism preserves null sets is needed. ∎

The result describes a change in normalized restart behavior while (5) holds on both sides of the transition. It does not supply a uniform bound on the finite entrance time over all parameters and initializations.

4 An explicit irrational restart word

The following fixed rational objective gives a concrete instance of the rotation regime.

Corollary 4.1. *Let*

$$H = \operatorname{diag}\left(\frac{79}{100}, \frac{41}{50}\right), \qquad x_0 = y_0 = (1, 1), \qquad s = 1.$$

Use the recurrence (2)–(3). *Then the restart intervals have irrational limiting frequencies*

$$\lim_{N\to\infty}\frac{N_4(N)}{N}=\frac{\log(343/216)}{\log 2},\qquad \lim_{N\to\infty}\frac{N-N_4(N)}{N}=\frac{\log(432/343)}{\log 2}.$$

The word is not eventually periodic. The normalized restart states have the exact signed limit set (7), *with*

$$A=\frac{216}{343},\quad B=\frac{432}{343},\quad R^2=\frac{2140663}{2904768}.$$

At the same time,

$$\|x_k\|\le\sqrt{2}\left(\frac{21}{100}\right)^{3\lfloor k/4\rfloor}.$$

Proof. Here $u=21/100$ and $v=9/50$ satisfy (1). Direct substitution gives the displayed constants and $B/A=2$, with $A<1<B$. The two accepted return matrices are

$$\operatorname{diag}(u^3,v^3)\quad\text{and}\quad\operatorname{diag}\left(\frac{u^3}{80},-\frac{v^3}{40}\right).$$

If $\log(343/216)/\log 2=p/q$ were rational with integers p and $q\ge 1$, then $(343/216)^q=2^p$. Comparing powers of the prime 7 is impossible. Thus the frequency is irrational, and all conclusions follow from Theorem 1. ∎

For this initialization the ratio begins inside the invariant interval: exact comparison gives $A^2R^2<1<B^2R^2$. The example does not rely on avoiding a switching equality, since the theorem covers such equalities. One can nevertheless check that no equality occurs here. The return rule gives $r_n=A^n2^{N_4(n)}$, whose square has only the primes 2, 3 and 7, whereas

$$R^2=\frac{7^3\,79^2}{2^6\,3^3\,41^2}.$$

This excludes $r_n=R$ for every n. The same prime comparison excludes visits to the two interval endpoints, even though all four signed endpoint states are subsequential limits.

The exact symbolic identities in (10)–(11) and the rational substitutions above are reproduced by the accompanying script. Infinite recurrence, frequency and limit-set conclusions follow from the proofs, not from a finite orbit computation. The example is consistent with the point-convergence results in [2]. The modal timing discussion in [1] does not assert eventual periodicity for this clock.

References

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